ENME608 Mechanisms

Cam Design (Norton Ch. 8)

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Contents

Overview of Cam–Follower Systems

A cam–follower converts continuous rotary input into a desired reciprocating or oscillating output motion through *direct contact*. Unlike a four-bar linkage, the motion program can be designed to arbitrary specification.

Equivalent four-bar linkage: At any instant, the cam–follower can be replaced by an equivalent four-bar linkage whose coupler length equals the radius of curvature of the cam at that point (instantaneous centre of curvature method).

Key variable definitions

Symbol	Meaning	Notes
θ	Cam rotation angle	Independent variable
β	Segment angle	Duration of one motion segment
h	Lift (stroke)	Total displacement of segment
s	Displacement	$s = s(\theta)$
v	Velocity	$v = ds/d\theta$ (w.r.t. θ, not time)
a	Acceleration	$a = d^2s/d\theta^2$
j	Jerk	$j = d^3s/d\theta^3$

Converting from θ-derivatives to time derivatives (cam speed ω = const):

$$\dot{s} = v\omega, \qquad \ddot{s} = a\omega^2, \qquad \dddot{s} = j\omega^3$$

SVAJ Diagrams and Motion Programs

SVAJ diagrams plot s, v, a, j against cam angle θ for a full 360Â° cycle. They are the primary design tool for checking continuity requirements.

Standard motion programs:

- **RF** — Rise then Fall (no dwell)
- **RFD** — Rise, Fall, Dwell (single dwell)
- **RDF** — Rise, Dwell, Fall (single dwell)
- **RDFD / DRDF** — Rise, Dwell, Fall, Dwell (*double dwell*)

Law of Cam Design

Exam focus. The Law of Cam Design is the central rule for all cam function selection.

> **The displacement s, velocity v, and acceleration a must be continuous at all points in the cam cycle. The jerk j must be finite everywhere.**

Discontinuities in a (infinite j) cause impact loads. Discontinuities in v or s are obviously unacceptable. This law governs which functions are permissible.

Consequence for dwell transitions: At the boundary between a dwell and a motion segment, $v = 0$ and $a = 0$ must hold *on the motion side* to match the dwell (where $v = 0$, $a = 0$). Any function that returns $a \neq 0$ at a dwell boundary violates the law.

Unacceptable Double-Dwell Functions

The following functions are **not suitable** for double-dwell motion because they violate the Law of Cam Design at dwell transitions:

Function	Problem	Violated condition
Constant velocity	a is infinite at boundaries	Jerk infinite
Constant acceleration	v is discontinuous at ends	Velocity discontinuous
Cubic displacement	a is discontinuous	Jerk infinite
Simple Harmonic (SHM)	$a \neq 0$ at dwell transitions	Violates law at dwell boundary

SHM: $s = \frac{h}{2}\left[1 - \cos\left(\frac{\pi\theta}{\beta}\right)\right]$ gives $a = \frac{h\pi^2}{2\beta^2}\cos\left(\frac{\pi\theta}{\beta}\right)$, which is non-zero at $\theta = 0$ and $\theta = \beta$, hence a finite jerk discontinuity exists at dwell boundaries.

Acceptable Double-Dwell Functions

Cycloidal

$$s = h\left[\frac{\theta}{\beta} - \frac{1}{2\pi}\sin\left(\frac{2\pi\theta}{\beta}\right)\right]$$

$$v = \frac{h}{\beta}\left[1 - \cos\left(\frac{2\pi\theta}{\beta}\right)\right], \quad a = \frac{2\pi h}{\beta^2}\sin\left(\frac{2\pi\theta}{\beta}\right), \quad j = \frac{4\pi^2 h}{\beta^3}\cos\left(\frac{2\pi\theta}{\beta}\right)$$

$v = 0$ and $a = 0$ at $\theta = 0$ and $\theta = \beta$. Fully satisfies the law. Peak acceleration coefficient $C_a = 2\pi \approx 6.283$.

4-5-6-7 Polynomial (Double Dwell)

Satisfies $s = 0$, $v = 0$, $a = 0$, $j = 0$ at $\theta = 0$ and $s = h$, $v = 0$, $a = 0$, $j = 0$ at $\theta = \beta$ (8 boundary conditions $\rightarrow$ degree 7):

$$s = h\left[35\left(\frac{\theta}{\beta}\right)^4 - 84\left(\frac{\theta}{\beta}\right)^5 + 70\left(\frac{\theta}{\beta}\right)^6 - 20\left(\frac{\theta}{\beta}\right)^7\right] \tag{1}$$

$$v = \frac{h}{\beta}\left[140\left(\frac{\theta}{\beta}\right)^3 - 420\left(\frac{\theta}{\beta}\right)^4 + 420\left(\frac{\theta}{\beta}\right)^5 - 140\left(\frac{\theta}{\beta}\right)^6\right] \tag{2}$$

$$a = \frac{h}{\beta^2}\left[420\left(\frac{\theta}{\beta}\right)^2 - 1680\left(\frac{\theta}{\beta}\right)^3 + 2100\left(\frac{\theta}{\beta}\right)^4 - 840\left(\frac{\theta}{\beta}\right)^5\right] \tag{3}$$

$$j = \frac{h}{\beta^3}\left[840\left(\frac{\theta}{\beta}\right) - 5040\left(\frac{\theta}{\beta}\right)^2 + 8400\left(\frac{\theta}{\beta}\right)^3 - 4200\left(\frac{\theta}{\beta}\right)^4\right] \tag{4}$$

3-4-5 Polynomial (Double Dwell)

6 boundary conditions (no jerk constraints): degree 5.

$$s = h\left[10\left(\frac{\theta}{\beta}\right)^3 - 15\left(\frac{\theta}{\beta}\right)^4 + 6\left(\frac{\theta}{\beta}\right)^5\right]$$

SCCA Family (Modified Trapezoidal and Modified Sine)

SCCA = Sine–Constant–Cosine–Acceleration. A family of functions blending sinusoidal and constant-acceleration segments. Defined in normalised coordinates $x = \theta/\beta$, $y = s/h$, with parameters $b + c + d = 1$ (zone fractions).

Key table:

Function	C_v (peak vel.)	C_a (peak accel.)	Best for
Modified Trapezoidal	2.000	**4.888**	**Minimum peak acceleration**
Modified Sine	**1.760**	5.528	**Minimum peak velocity**
Cycloidal	2.000	6.283	Zero a at all boundaries

Exam focus. Modified Trapezoidal → minimum acceleration. Modified Sine → minimum velocity. Know which to choose.

Modified Sine parameters: $b = 0.25$, $c = 0$, $d = 0.75$, $C_a = 5.528$.

Modified Trapezoidal parameters: $b = 0.25$, $c = 0.5$, $d = 0.25$, $C_a = 4.888$.

Single-Dwell Cam Design

For RFD or RDF motion, rise and fall together form **one continuous segment** described by a single polynomial. The key constraint is that *velocity must be zero at the rise–fall boundary* to prevent follower overshoot (the follower must momentarily stop at the peak before reversing).

Do *not* use two separate double-dwell functions joined at the peak — the acceleration would return to zero at the boundary, creating a finite jerk spike.

Procedure:

1. Treat the combined rise+fall as one segment: $\theta \in [0,\ \beta_1 + \beta_2]$.
2. Apply boundary conditions at $\theta = 0$ (start), $\theta = \beta_1$ (peak), and $\theta = \beta_1 + \beta_2$ (end).
3. Solve for polynomial coefficients.

At $\theta = \beta_1$ (peak): $s = h$, $v = 0$. The acceleration at the peak is *non-zero* and *negative* (follower decelerating). This is correct and expected.

Non-Zero Initial Cam Angle

Exam focus. Always state the domain of θ for each segment. Marks are awarded for this.

If a segment does not start at $\theta = 0$, use the substitution $\theta' = \theta - \alpha$ where α is the starting angle of the segment. The function is then written in terms of θ', valid over $\theta' \in [0, \beta]$ (equivalently $\theta \in [\alpha, \alpha + \beta]$).

Cam Sizing

Key circles

- **Base circle** (radius R_b): smallest circle on cam body.

- **Prime circle** (radius R_p): $R_p = R_b + r_{\text{roller}}$ for roller followers. For flat-face followers, $R_p = R_b$.

Pressure angle ϕ

The angle between the contact force direction and the follower velocity direction.

$$\tan\phi = \frac{v - \varepsilon\,\omega}{s + R_p} \cdot \frac{1}{\omega}$$

where ε is eccentricity (offset). Must satisfy $\phi < 30°$ (translating) or $\phi < 35$–$40°$ (oscillating). Increase R_b to reduce ϕ.

Radius of curvature

For roller followers: roller radius $< \rho_{\min}$ (minimum radius of curvature of pitch curve), otherwise the follower cannot follow the cam profile (undercutting).

For flat-face followers: flat-face width $\geq v_{\max} - v_{\min}$ (ensure face always contacts cam).

Tutorial 7 — Worked Solutions

Problems 8-1 and 8-4

*8-1 Figure P8-1 shows the cam and follower from Problem 6-65. Using graphical methods, find and sketch the equivalent fourbar linkage for this position of the cam and follower.

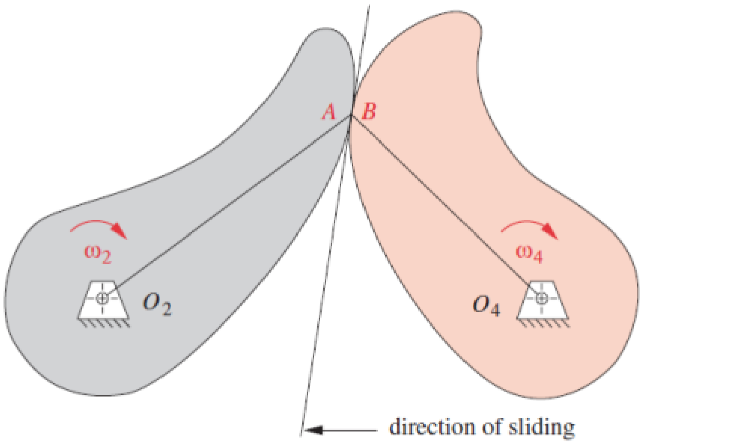

FIGURE P8-1

Problem 8-1 statement and figure.

*8-4 Figure P8-2 shows a cam and follower. Using graphical methods, find the pressure angle at the position shown.

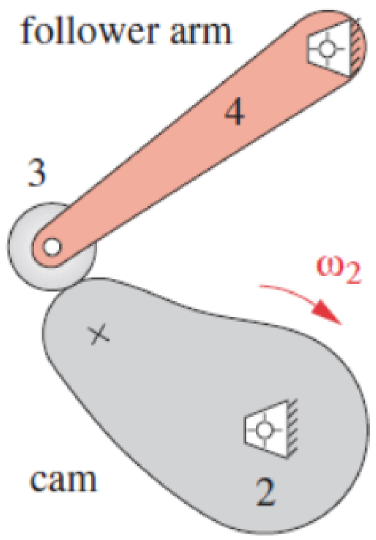

FIGURE P8-2

Problem 8-4 statement and figure.

These are graphical problems (equivalent four-bar linkage and pressure angle by graphical construction). Refer to lecture slides for procedure. For 8-4: the pressure angle is measured between the common normal at the contact point and the tangent to the follower path.

8-1 answer: pressure angle $\phi = \boxed{4.9^\circ}$ (graphical construction).

8-4 $\varphi = 4.9°$.

Problem 8-4 worked solution (pressure-angle construction).

Problem 8-46 — 4-5-6-7 Polynomial, Double Dwell

‡8-46 Design a double-dwell cam to move a follower from 0 to 50 mm in 75°, dwell for 75°, fall 50 mm in 75°, and dwell for the remainder. The total cycle must take 5 sec. Use a 4-5-6-7 polynomial function for rise and fall and plot the *s v a j* diagrams.

Problem 8-46 statement.

Problem statement: Design a double-dwell cam to move a follower from 0 to 50 mm in 75°, dwell for 75°, fall 50 mm in 75°, and dwell for the remainder. Use a 4-5-6-7 polynomial for rise and fall.

Motion program: DRDF (double-dwell, rise–dwell–fall–dwell).

Segment summary:

Segment	β (deg)	β (rad)	h	Domain θ
Rise	$\beta_1 = 75°$	$\frac{5\pi}{12}$	50 mm	$0 \le \theta \le 75°$
Dwell (high)	$\beta_2 = 75°$	$\frac{5\pi}{12}$	—	$75° \le \theta \le 150°$
Fall	$\beta_3 = 75°$	$\frac{5\pi}{12}$	50 mm	$150° \le \theta \le 225°$
Dwell (low)	$\beta_4 = 135°$	$\frac{3\pi}{4}$	—	$225° \le \theta \le 360°$

Rise segment ($0 \le \theta \le \beta_1 = \frac{5\pi}{12}$ rad, $h = 50$ mm):

Using the 4-5-6-7 polynomial with $\beta = \beta_1$:

$$s_1 = 50\left[35\left(\frac{\theta}{\beta_1}\right)^4 - 84\left(\frac{\theta}{\beta_1}\right)^5 + 70\left(\frac{\theta}{\beta_1}\right)^6 - 20\left(\frac{\theta}{\beta_1}\right)^7\right] \text{ mm}$$

$$v_1 = \frac{50}{\beta_1}\left[140\left(\frac{\theta}{\beta_1}\right)^3 - 420\left(\frac{\theta}{\beta_1}\right)^4 + 420\left(\frac{\theta}{\beta_1}\right)^5 - 140\left(\frac{\theta}{\beta_1}\right)^6\right]$$

$$a_1 = \frac{50}{\beta_1^2}\left[420\left(\frac{\theta}{\beta_1}\right)^2 - 1680\left(\frac{\theta}{\beta_1}\right)^3 + 2100\left(\frac{\theta}{\beta_1}\right)^4 - 840\left(\frac{\theta}{\beta_1}\right)^5\right]$$

$$j_1 = \frac{50}{\beta_1^3}\left[840\left(\frac{\theta}{\beta_1}\right) - 5040\left(\frac{\theta}{\beta_1}\right)^2 + 8400\left(\frac{\theta}{\beta_1}\right)^3 - 4200\left(\frac{\theta}{\beta_1}\right)^4\right]$$

At $\theta = 0$: $s_1 = 0$, $v_1 = 0$, $a_1 = 0$, $j_1 = 0$. ✓
At $\theta = \beta_1$: $s_1 = 50$ mm, $v_1 = 0$, $a_1 = 0$, $j_1 = 0$. ✓

High dwell ($75° \le \theta \le 150°$):

$$s_2 = h = 50\,\text{mm}, \quad v_2 = 0, \quad a_2 = 0, \quad j_2 = 0$$

Fall segment ($150° \le \theta \le 225°$, $\beta_3 = \frac{5\pi}{12}$, $h = 50\,\text{mm}$):

Let $\theta' = \theta - (\beta_1 + \beta_2) = \theta - 150°$ so that $\theta' \in [0, \beta_3]$.

The displacement *decreases* from h to 0, so:

$$s_3 = h - s(\theta',\, \beta_3,\, h) = 50\left[1 - 35\left(\frac{\theta'}{\beta_3}\right)^4 + 84\left(\frac{\theta'}{\beta_3}\right)^5 - 70\left(\frac{\theta'}{\beta_3}\right)^6 + 20\left(\frac{\theta'}{\beta_3}\right)^7\right] \text{ mm}$$

$$v_3 = -v(\theta', \beta_3, h), \quad a_3 = -a(\theta', \beta_3, h), \quad j_3 = -j(\theta', \beta_3, h)$$

Domain stated as: $\theta \in [150°,\, 225°]$, i.e. $\theta' = \theta - \frac{150\pi}{180}$.

Low dwell ($225° \le \theta \le 360°$):

$$s_4 = 0, \quad v_4 = 0, \quad a_4 = 0, \quad j_4 = 0$$

Continuity check at all boundaries:

- $\theta = 0$: $s = 0$, $v = 0$, $a = 0$, $j = 0$ (all zero by 4-5-6-7 B.C.) ✓
- $\theta = 75°$: Rise gives $s = h$, $v = 0$, $a = 0$, $j = 0$; Dwell gives same. ✓
- $\theta = 150°$: High dwell gives $s = h$, $v = 0$, $a = 0$; Fall start gives $s = h$, $v = 0$, $a = 0$. ✓
- $\theta = 225°$: Fall gives $s = 0$, $v = 0$, $a = 0$; Low dwell gives same. ✓

Law of Cam Design satisfied everywhere. ✓

Problem 8-9 — Single-Dwell Polynomial

‡8-9 Design a single-dwell cam to move a follower from 0 to 2" in 60°, fall 2" in 90°, and dwell for the remainder. The total cycle must take 2 sec. Choose suitable functions for rise and fall to minimize accelerations. Plot the *s v a j* diagrams.

Problem 8-9 statement.

Problem statement: Design a single-dwell cam to move a follower from 0 to 2 in in 60°, fall 2 in in 90°, and dwell for the remainder (210°). Total cycle 2 sec. Choose functions to minimise acceleration.

Motion program: RFD (single dwell at bottom).

Strategy: Since this is a single-dwell case, the rise and fall must be combined into one continuous polynomial over the total angle $\beta = \beta_1 + \beta_2 = 60° + 90° = 150° = \frac{5\pi}{6}$ rad. A separate function for rise and fall would create an acceleration discontinuity at the peak.

Set $v = 0$ at $\theta = \beta_1$ (peak) to avoid overshoot of displacement.

Boundary conditions:

Let $A = \beta_1/\beta = 60/150 = 0.4$.

Boundary	Condition	Equation
$\theta = 0$	$s = 0$, $v = 0$, $a = 0$	$C_0 = C_1 = C_2 = 0$
$\theta = \beta_1$	$s = h = 2\,\text{in}$	Eq. (1)
$\theta = \beta_1$	$v = 0$ (no overshoot)	Eq. (2)
$\theta = \beta$	$s = 0$	Eq. (3)
$\theta = \beta$	$v = 0$	Eq. (4)
$\theta = \beta$	$a = 0$	Eq. (5)

With $C_0 = C_1 = C_2 = 0$, the polynomial is:

$$s(\theta) = C_3\left(\frac{\theta}{\beta}\right)^3 + C_4\left(\frac{\theta}{\beta}\right)^4 + C_5\left(\frac{\theta}{\beta}\right)^5 + C_6\left(\frac{\theta}{\beta}\right)^6 + C_7\left(\frac{\theta}{\beta}\right)^7$$

Setting up the 5-equation system (with $A = 0.4$, $h = 2\,\text{in}$):

From Eq. (1) [$s(\beta_1) = h$]:

$$C_3A^3 + C_4A^4 + C_5A^5 + C_6A^6 + C_7A^7 = h$$

From Eq. (2) [$v(\beta_1) = 0$, i.e. $ds/d(\theta/\beta) = 0$ at $\theta/\beta = A$]:

$$3C_3A^2 + 4C_4A^3 + 5C_5A^4 + 6C_6A^5 + 7C_7A^6 = 0$$

From Eq. (3) [$s(\beta) = 0$, i.e. at $\theta/\beta = 1$]:

$$C_3 + C_4 + C_5 + C_6 + C_7 = 0$$

From Eq. (4) [$v(\beta) = 0$]:

$$3C_3 + 4C_4 + 5C_5 + 6C_6 + 7C_7 = 0$$

From Eq. (5) [$a(\beta) = 0$]:

$$6C_3 + 12C_4 + 20C_5 + 30C_6 + 42C_7 = 0$$

Matrix form:

$$\begin{bmatrix} A^3 & A^4 & A^5 & A^6 & A^7 \\ 3A^2 & 4A^3 & 5A^4 & 6A^5 & 7A^6 \\ 1 & 1 & 1 & 1 & 1 \\ 3 & 4 & 5 & 6 & 7 \\ 6 & 12 & 20 & 30 & 42 \end{bmatrix} \begin{bmatrix} C_3 \\ C_4 \\ C_5 \\ C_6 \\ C_7 \end{bmatrix} = \begin{bmatrix} h \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Substituting $A = 0.4$, $h = 2\,\text{in}$ and solving via matrix inversion:

$$\boxed{C_3 = 289.352, \quad C_4 = -1229.745, \quad C_5 = 1953.125, \quad C_6 = -1374.421, \quad C_7 = 361.690}$$

Final displacement function for $\theta \in [0°, 150°]$:

$$\begin{aligned} s(\theta) = 289.352\left(\frac{\theta}{\beta}\right)^3 &- 1229.745\left(\frac{\theta}{\beta}\right)^4 + 1953.125\left(\frac{\theta}{\beta}\right)^5 \\ &- 1374.421\left(\frac{\theta}{\beta}\right)^6 + 361.690\left(\frac{\theta}{\beta}\right)^7 \quad [\text{in}] \end{aligned}$$

with $\beta = 150° = \frac{5\pi}{6}$ rad.

Dwell segment ($150° \le \theta \le 360°$):

$$s = 0, \quad v = 0, \quad a = 0, \quad j = 0$$

Problem 8-8 — Modified Sinusoidal SCCA, Double Dwell

‡8-8 Design a double-dwell cam to move a follower from 0 to 1.5" in 45°, dwell for 150°, fall 1.5" in 90°, and dwell for the remainder. The total cycle must take 6 sec. Choose suitable functions for rise and fall to minimize velocities. Plot the *s v a j* diagrams.

Problem 8-8 statement.

Problem statement: Design a double-dwell cam: rise 1.5 in in 45°, dwell 150°, fall 1.5 in in 90°, dwell 75° (total 360°). Choose functions to *minimise velocities.*

Motion program: RDFD.

Function selection: To minimise peak velocity, use **Modified Sinusoidal (Modified Sine)** SCCA, which gives $C_v = 1.760$ (lowest of all standard double-dwell functions).

SCCA parameters for Modified Sine:

$$b = 0.25, \quad c = 0, \quad d = 0.75, \quad C_a = 5.528$$

(with $b + c + d = 1$).

Segment summary:

Segment	β	h
Rise	$\beta_1 = 45° = \pi/4$	$h_1 = 1.5\,\text{in}$
High dwell	$\beta_2 = 150° = 5\pi/6$	$h_2 = 0$ (constant at h_1)
Fall	$\beta_3 = 90° = \pi/2$	$h_3 = 1.5\,\text{in}$
Low dwell	$\beta_4 = 75° = 5\pi/12$	$h_4 = 0$ (constant at 0)

Modified Sinusoidal SCCA rise function in normalised coordinates $x = \theta/\beta$, $y = s/h$:

The SCCA function is defined over 5 zones (using $b = 0.25$, $c = 0$, $d = 0.75$):

Zone 1 ($0 \le x \le b/2 = 0.125$):

$$y_1 = C_a\left[\frac{b}{4\pi}x - \left(\frac{b}{4\pi}\right)^2 \sin\left(\frac{4\pi x}{b}\right)\right] = C_a\left[\frac{b}{\pi}x - \left(\frac{b}{\pi}\right)^2 \sin\left(\frac{\pi x}{b/2}\right)\cdot\frac{1}{4}\right]$$

More precisely, for Modified Sine, Zone 1 ($0 \le x \le b/2$):

$$y_1 = C_a\left[\frac{b}{4\pi}\left(\frac{2\pi x}{b}\right) - \frac{b}{4\pi}\sin\left(\frac{2\pi x}{b}\right)\right] = C_a\left[\frac{x}{2} - \frac{b}{4\pi}\sin\left(\frac{2\pi x}{b}\right)\right]$$

Zone 2 ($b/2 \le x \le (1-d)/2 = 0.125$): Zones 1 and 2 coincide for $c = 0$.

Zone 3 ($0.125 \le x \le 0.875$, i.e. $(1-d)/2 \le x \le (1+d)/2$):

$$y_3 = C_a\left[\left(\frac{b}{2\pi} + \frac{c}{2}\right)x + d^2\left(\frac{1}{8} - \frac{1}{\pi^2}\right) - \frac{d^2}{\pi^2}\cos\left(\frac{\pi(x-(1-d)/2)}{d}\right)\right]$$

In practice, the full substitution back to physical coordinates proceeds via:

$$s = h \cdot y\left(\frac{\theta}{\beta}\right), \quad \theta \in [0, \beta_1]$$

Rise ($\theta \in [0°, 45°]$): Substitute $x = \theta/\beta_1$ into the Modified Sine SCCA formulae above with $h = 1.5\,\text{in}$, $\beta = \pi/4$.

High dwell ($45° \le \theta \le 195°$):

$$s = h = 1.5\,\text{in}, \quad v = 0, \quad a = 0$$

Fall ($195° \le \theta \le 285°$): Let $\theta' = \theta - 195°$. The fall is the mirror of the rise: $y' = 1 - y(\theta'/\beta_3)$, with $h = 1.5\,\text{in}$, $\beta_3 = 90° = \pi/2$.

$$s_{\text{fall}} = h\left[1 - y\left(\frac{\theta'}{\beta_3}\right)\right]$$

Low dwell ($285° \le \theta \le 360°$):

$$s = 0, \quad v = 0, \quad a = 0$$

Continuity check: The Modified Sine function by construction gives $y = 0$, $dy/dx = 0$ at $x = 0$ and $y = 1$, $dy/dx = 0$ at $x = 1$, with zero acceleration at both endpoints. All dwell transitions are therefore continuous in s, v, a. ✓

Minimum velocity confirmation: $C_v = 1.760$ compared to Modified Trapezoidal ($C_v = 2.000$) and Cycloidal ($C_v = 2.000$). Modified Sine is the correct choice. ✓